

Methodology

The sample consists of 103 E+A post-starburst galaxies that were filtered from a dataset (using SQL) from the Vera C. Rubin Observatory LSST photometric catalog. Our analysis was used in Python 3.12, where the galaxy catalog was read from a CSV file using Python's `csv` module and extracting the composite model flux (`cModelFlux`) across 6 different photometric bands(u,g,r,i,z,y) and the 2nd order shape moments (`shape_xx` and `shape_yy`) for each one of the 103 E+A's. The CAS parameters were implemented as 3 different functions, **Concentration** being the first, where we computed the flux-weighted radii approximated from the shape moments, where `r_80,20` were scaled using a normalised band flux fraction to ensure that we were physically varying values across each sample. The **Asymmetry** was computed by comparing each galaxy's flux to its mirror position in the reversed band flux array. And finally, the **Smoothness** was computed by applying '`scipy.ndimage.uniform_filter`' to the flux array with a kernel size of 0.3 of the approximated Petrosian radius and a min kernel size of 3, enforced to prevent any galaxies from returning any null results. We then stored our results in a pandas DataFrame and exported and time-stamped our CSV file using '`dataFrame.to_csv()`'.